

Turtlesim & system inspection — quick reference

ROS 2 Fundamentals · Module 4 · Jazzy Jalisco on Ubuntu 24.04

START THINGS

<code>ros2 pkg executables turtlesim</code>	What can I run from this package?
<code>ros2 run turtlesim turtlesim_node</code>	Start the simulator (opens a window)
<code>ros2 run turtlesim turtle_teleop_key</code>	Drive it with the arrow keys
<code>Ctrl+C</code>	Stop whatever is running in this terminal

INSPECT A RUNNING SYSTEM

<code>ros2 node list</code>	The separate programs running now → M5
<code>ros2 topic list</code>	Continuous streams of data → M6
<code>ros2 service list</code>	Ask-and-get-an-answer requests → M7
<code>ros2 action list</code>	Long-running, cancellable goals → M8
<code>ros2 topic echo /turtle1/cmd_vel</code>	Watch data actually moving, live

WHEN NOTHING MOVES

1. Is anything being sent at all?

`ros2 topic echo /turtle1/cmd_vel`
Press the arrow keys and watch. Silence means the problem is on the **publishing** side — don't start by blaming the simulator.

2. Which window has your keyboard?

`turtle_teleop_key` reads keystrokes from *its own terminal*. If the Turtlesim window is focused, the keys go nowhere. Click the teleop terminal.

3. Is this terminal sourced?

`echo $ROS_DISTRO` → should print **jazzy**.
Empty means this shell doesn't know about ROS 2 yet (Module 3, Lesson 3).

WHEN THE WINDOW NEVER APPEARS

Almost never a ROS 2 problem. If the terminal logs look healthy but nothing opens, it's your display:

- `echo $DISPLAY` — empty is the tell
- **WSL2:** needs WSLg (Windows 11, or an X server)
- **VM:** guest additions + 3D acceleration enabled
- **SSH:** connect with `-X` for forwarding

Note on output: any terminal output shown in this course is illustrative. Exact package counts, node ids, point-release numbers and paths differ legitimately between machines — match the *shape* of the output, not the characters. · Every command here works unchanged on a real robot; `/cmd_vel` is the standard velocity interface, not a Turtlesim invention.